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BIOPROCESS

# *Understanding Planova™ Virus Filtration and Designing Robust Virus Filtration Processes*

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# Overview

# Line-Up of Planova Products

- **Pioneering nanofiltration technology since 1989**

| 1989 | 1991 | 2001 | 2009 |
|---|---|---|---|
| Planova 35N | Planova 15N | Planova 20N | Planova BioEX |
| HIV HCV | B19 MVM | B19 MVM | B19 MVM |
| Cellulose | Cellulose | Cellulose | PVDF |

# Planova Filters

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| **Filter** | **Planova N** | **Planova BioEX** |
|---|---|---|
| **Hollow fibers** | Regenerated cellulose | PVDF |
| **Pore size (nm)** | 15, 20, 35, (75) | Parvovirus removal |
| **Filtration surface ($m^2$)** | 4.0, 1.0, 0.3, 0.12, 0.01, 0.001 | 4.0, 1.0, 0.1, 0.01, 0.001, 0.0003 |
| **Max TMP (kPa)** | 98 | 343 |
| **Water flux (LMH)** | **20N**: 66 @ 98 kPa | 170 @ 343 kPa |
| **SIP (Autoclaving)** | NO | YES |
| **Post-use integrity test** | Leakage test<br>+ Gold particle test | Leakage test |

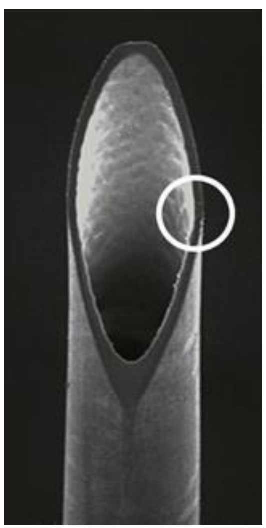
# Virus Removal Mechanism

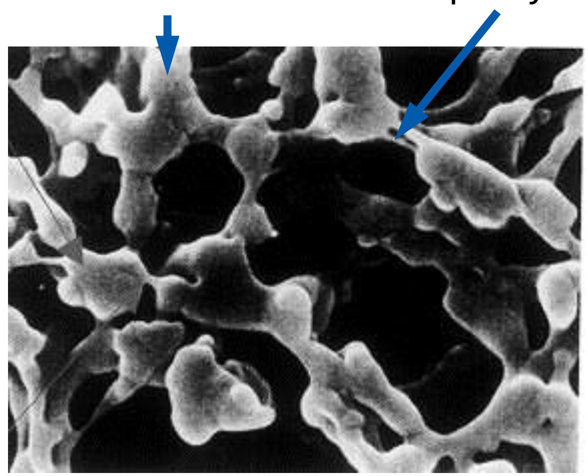
- Viruses bigger than the pore size of the filter are removed.

➡ **Size exclusion mechanism**

- Voids are connected 3-dimensionally with several capillaries.
- This structure is repeated through the membrane thickness.

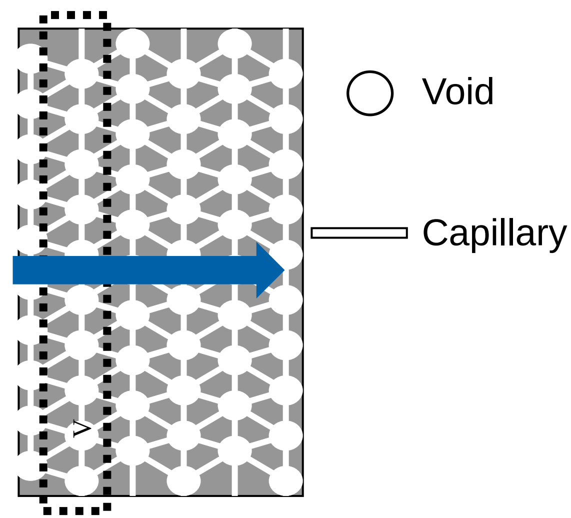
➡ **Multi-layer filtration mechanism**






Negative image

Multi-layer image



# Mechanism: Size Exclusion

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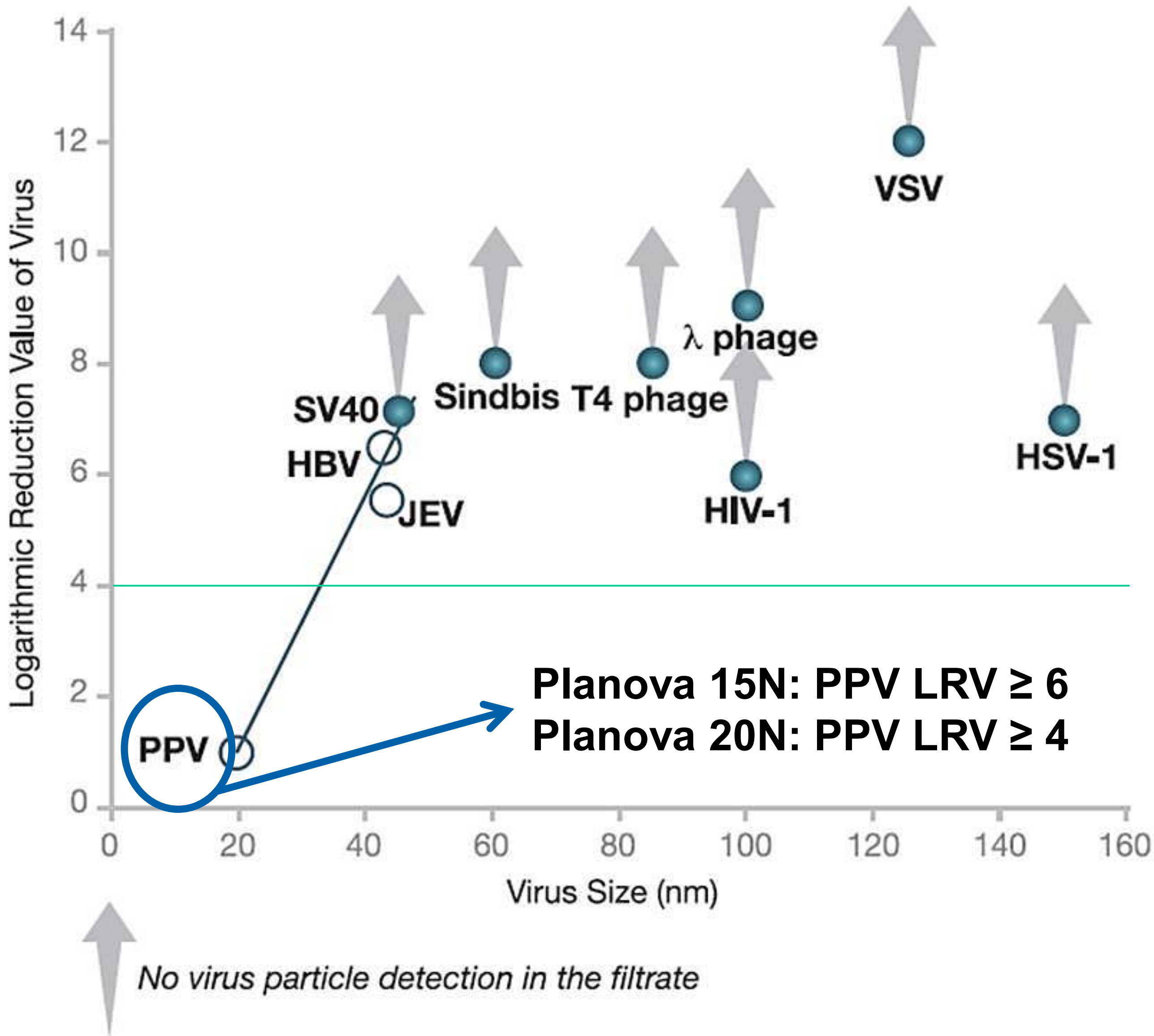
- Correlation between virus size and removal on Planova 35N



# Designed Pore Size Gradient Membrane

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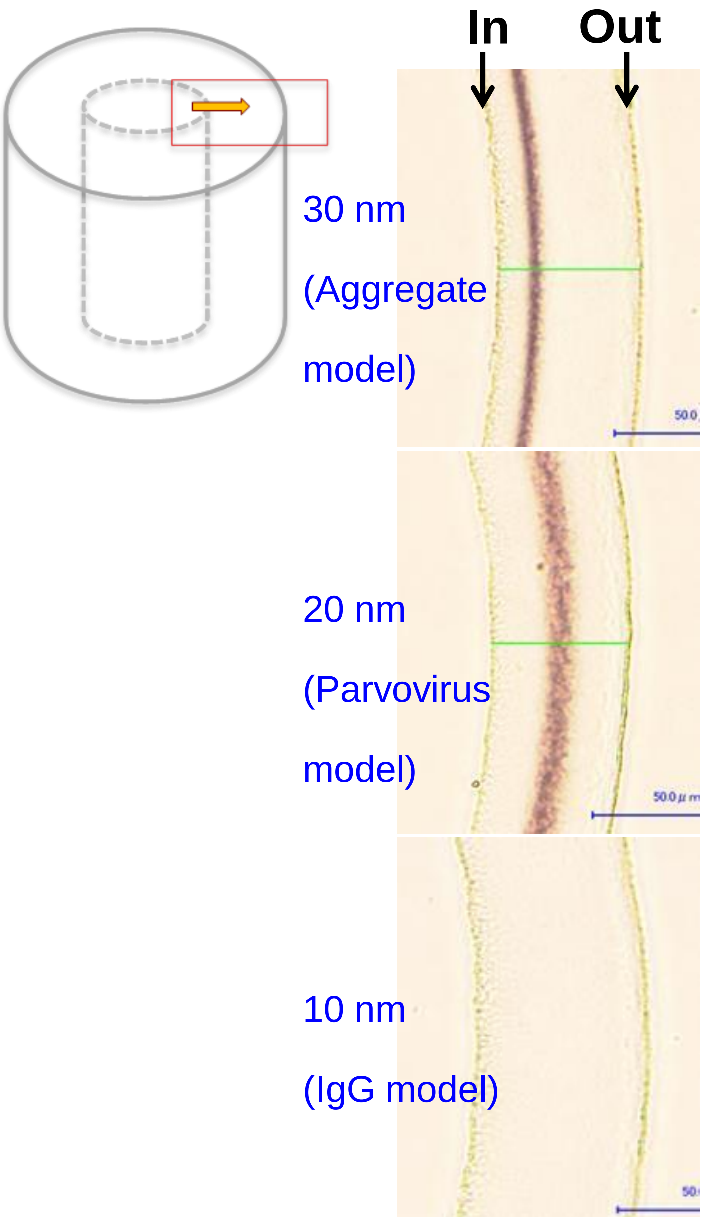


Planova 20N membrane
Model gold particles: 10, 20, 30 nm
Microscopic analysis

- 30 nm gold particle trapped at inner side as a function of pre-filter layer
- 20 nm gold particle trapped at center of membrane

⇒High capacity

- Resistance to plugging
- Stable flux performance
- High virus trapping capability

# Parvovirus Trapping Images

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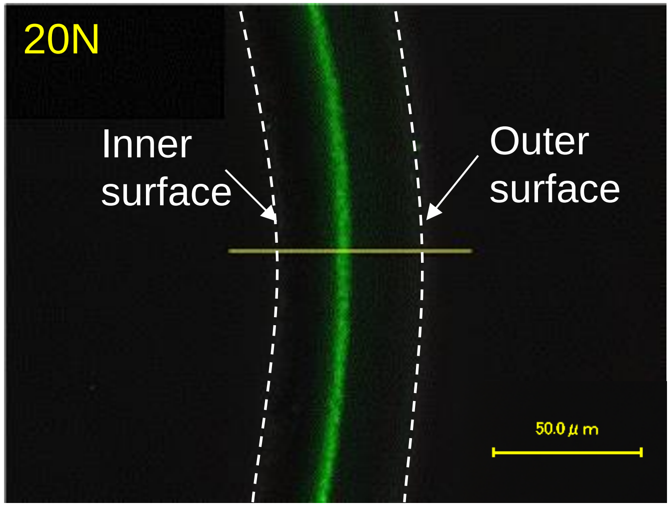
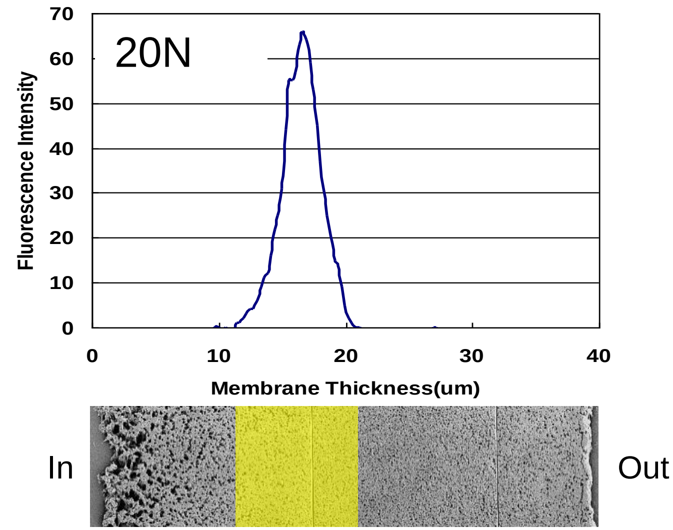
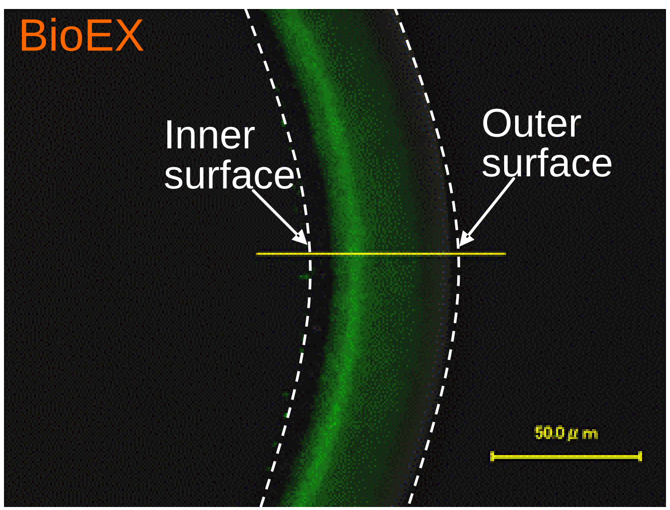
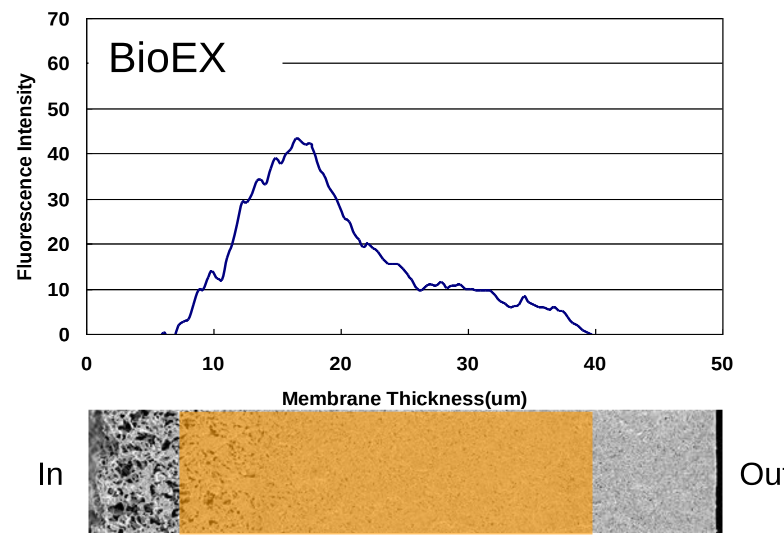


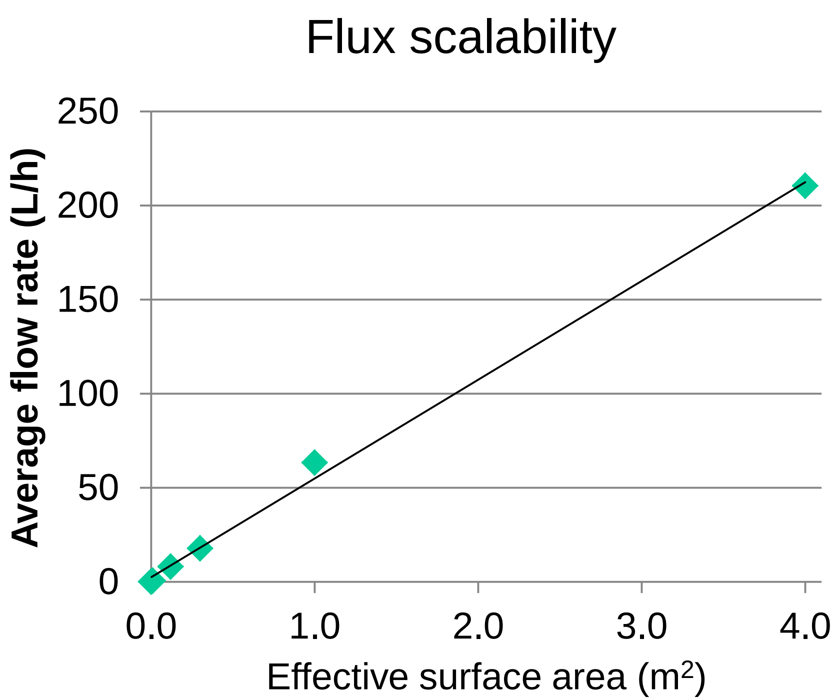



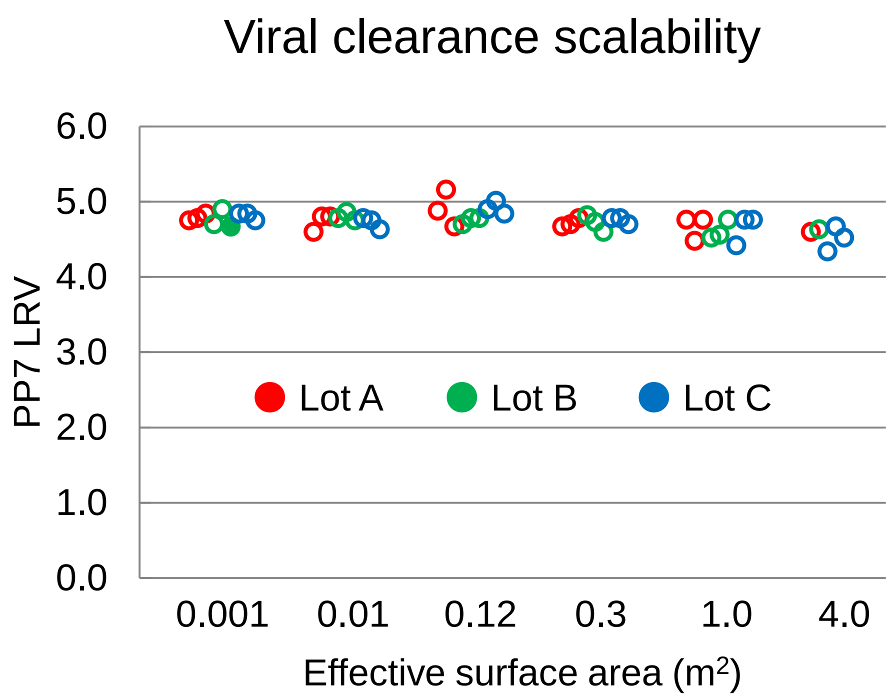




5 mg/mL IgG/100 mM NaCl (pH 4.5)

*Tomoko Hongo-Hirasaki, Asahi Kasei Medical, 2016 Planova Workshop* (adapted)

# Planova 20N Scalability




- Robust performance and viral clearance was observed for Planova 20N filters at all scales

*Buesing, et al., 2015 Planova Workshop, Athens, Greece*

# Applicable to Wide Range of Proteins

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| Pharmaceutical | MW (kDa) | P15N | P20N/PBioEX | P35N |
|---|---|---|---|---|
| Interleukin | 12 – 35 | | | |
| Interferon, GCS-F | 20 | | | |
| MCS-F, EPO, Thrombin | 30 – 40 | | | |
| Urokinase | 54 | | | |
| α1-protease inhibitor | 56 | | | |
| Factor-IX | 58 | | | |
| Protein C | 62 | | | |
| Hemoglobin, antithrombin | 65 | | | |
| Albumin | 66 | | | |
| Osteopontin | 70 | | | |
| Monoclonal IgG, Polyclonal IgG | 160 | | | |
| Factor XI | 180 | | | |
| Factor VIII | 300 | | | |
| Fibrinogen | 340 | | | |
| IgM | 800 | | | |

# Involvement of Virus Filter in Drug Development

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Preclinical development → Phase I → Phase II → Phase III

## Easy to platform

### Easy optimization

Stable filtration
Reliable virus removal
Time saving
Multi-product applicability

### Scalability

Reliable scalability
Reliable virus removal
Time and cost saving

### Consistency

Stable and reliable filtration
Assured virus safety
Time and cost saving

## Robust performance

Most important feature of virus filter

# Overview

1. Introduction
2. Findings for Planova Virus Filtration
   2.1. Key Factors of Robust Virus Filtration Processes
   2.2. Design Space for Robust and Stable Performance
3. Findings for Appropriate Viral Clearance Studies
4. Conclusions

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# What is Robust Virus Filtration?

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✓ Filterability
- Stable flux
- Small flux decay
- Predictable throughput filtration to target processing time
- Scalability; consistent performance across filter sizes

✓ Virus Removal
- Maintain virus LRV >4
- Reliable virus removal rate
- Scalability; consistent performance across filter sizes

# Factors Necessary for Robust Virus Filtration

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✓ Process parameters

- Protein conc.
- Solution property

  pH, ionic strength etc.
- Temperature
- Filtration pressure (flow rate)

✓ Filtration scale

- Lab to manufacturing

  Scalability, consistency

# What is Robust Filterability?

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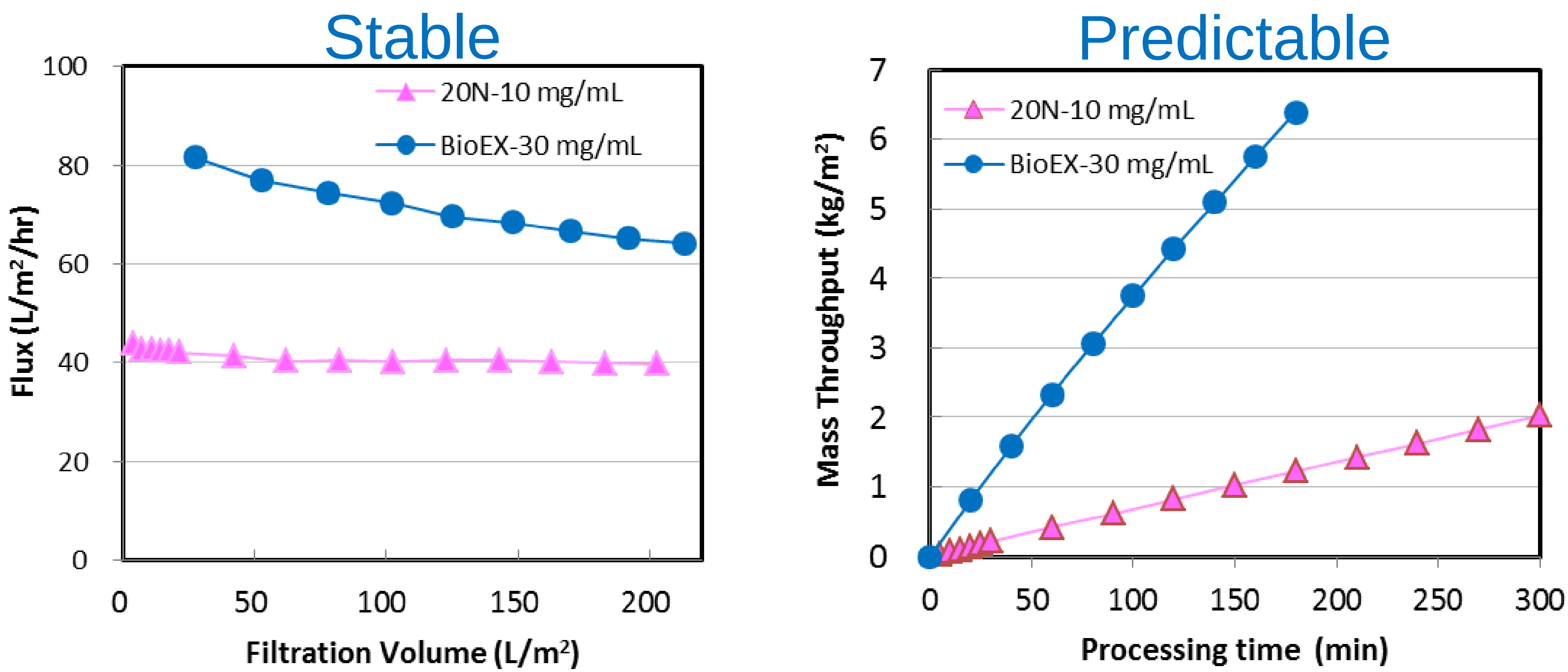


**Pressure**: 0.8 bar (P20N), 3.0 bar (PBioEX)

**Product solution**: 10 or 30 mg/mL IgG, 100 mM NaCl, pH 4.5

✓ Planova filters show stable flux with relatively small flux decay and predictable throughput for the duration of the filtration process.

# What is Robust Filterability: Impact of IgG Concentration on Flux Decay

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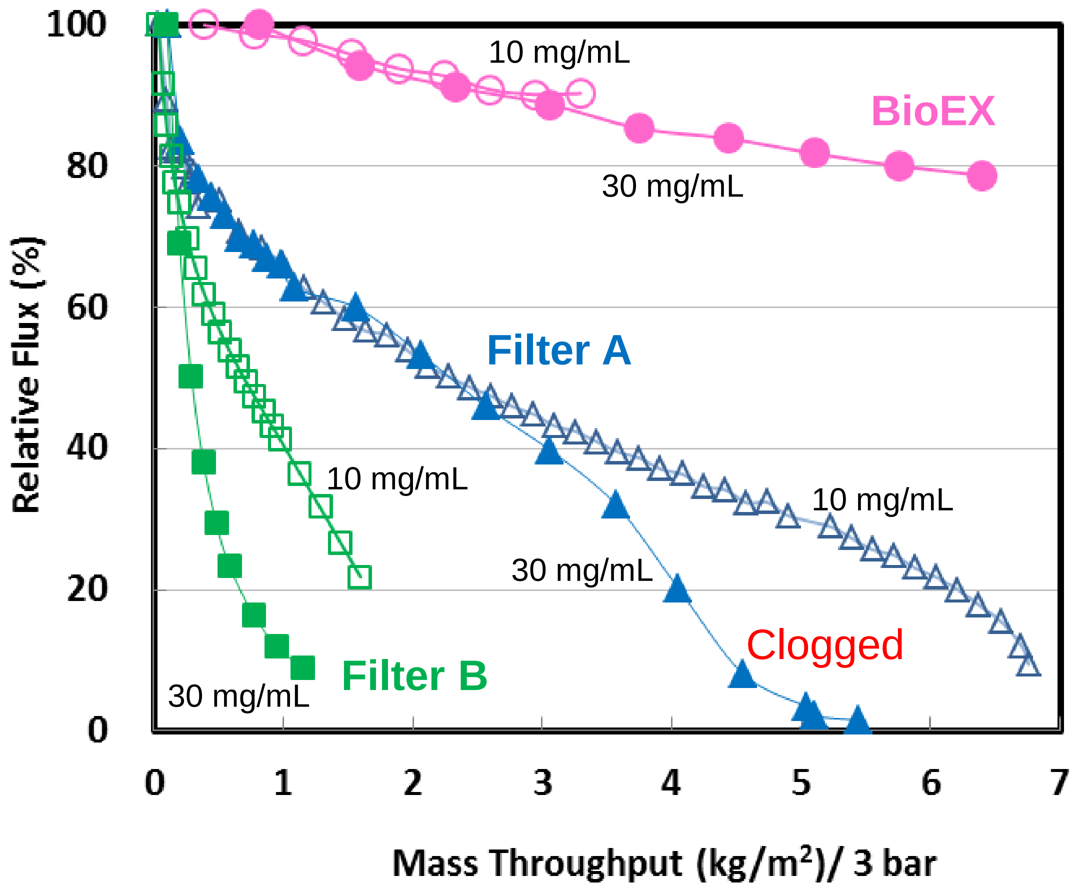
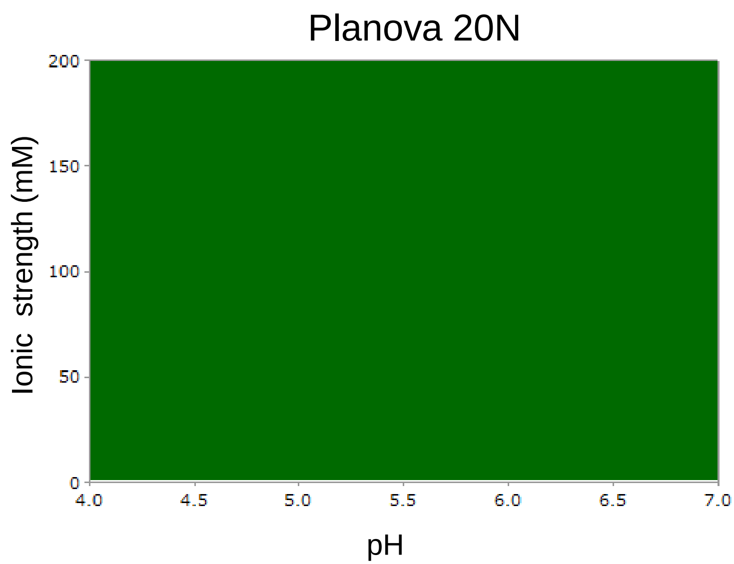
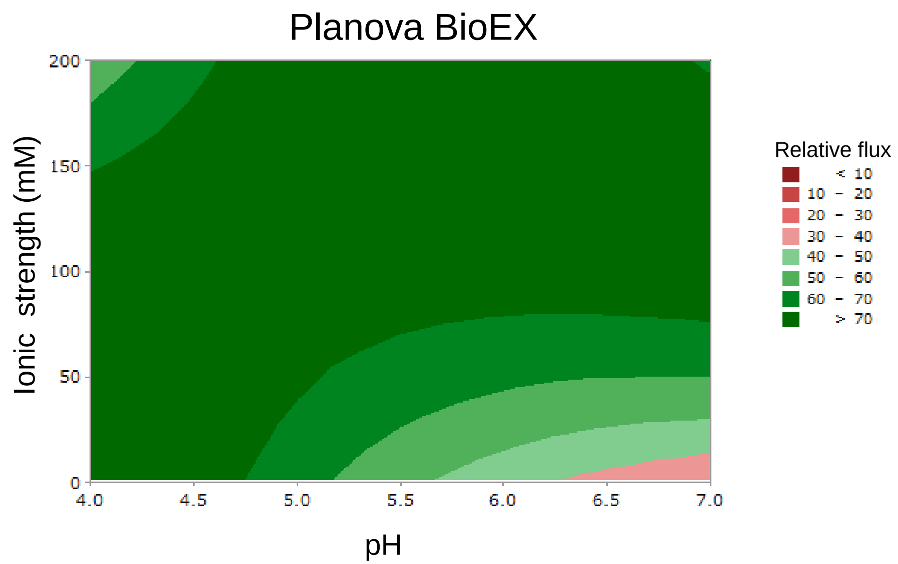
**Product solution**: 10 or 30 mg/mL IgG, 100 mM NaCl, pH 4 – 4.6

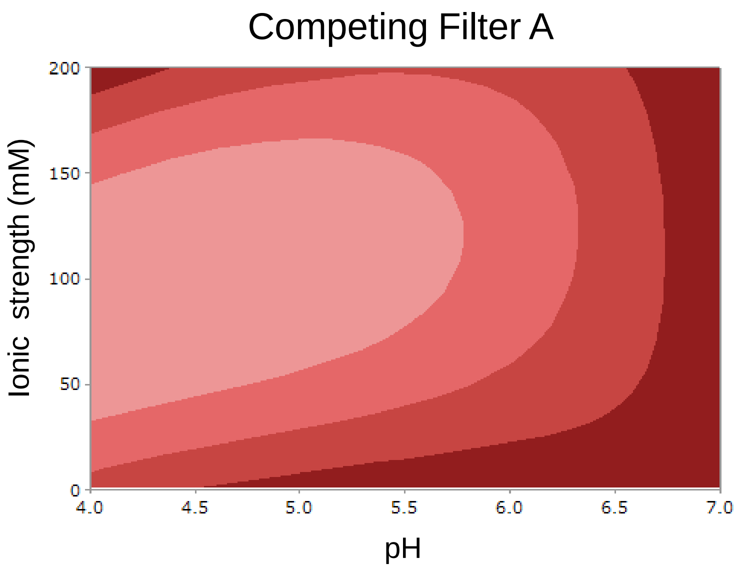



✓ Planova BioEX shows low flux decay even for mid to high IgG concentration.

# What is Robust Filterability:
# Impact of pH and ionic strength on flux decay




10 mg/mL human IgG, 3 h filtration

Relative flux (%)
= (3h flux / Initial flux) x 100

✓ Planova 20N and BioEX show stable flux over wide range of pH and ionic strength

# What is Robust Filterability: Impact of Multimer on Flux Decay

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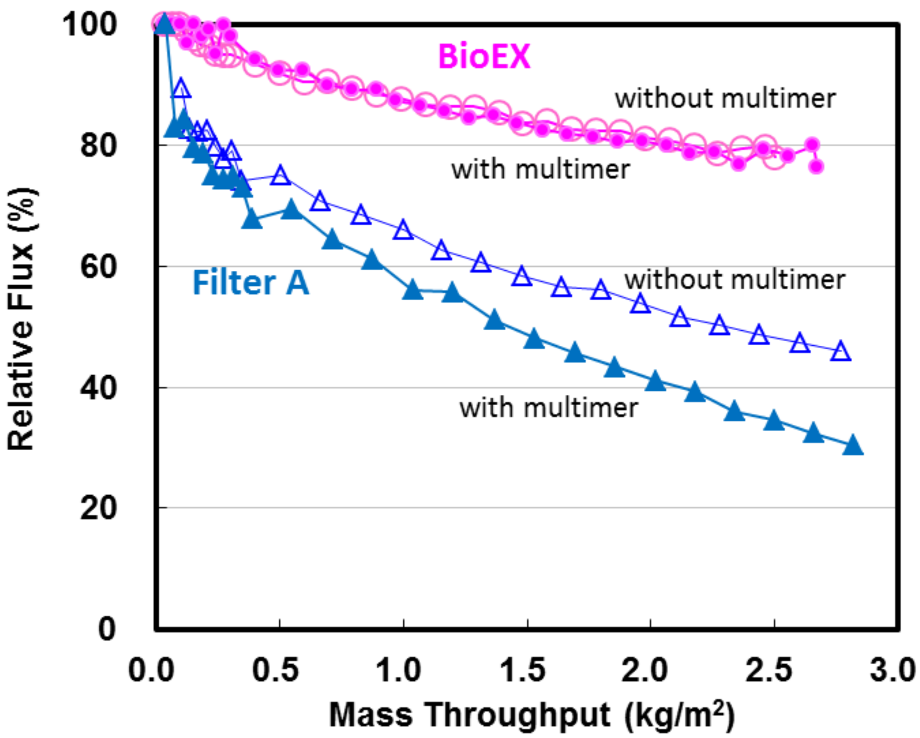
**Pressure**: 3.0 bar

**Product solution**: 10 mg/mL IgG, 100 mM NaCl, pH 4.5, 0.0025% multimer spiked (low pH denaturation)

**Prefilter**: Planova™35N (post-multimer spiking)




- ✓ Planova BioEX is less impacted by aggregates and shows lower flux decay compared to Filter A.
- ✓ Multimer caused by changes in parameters may impact filterability

# Robust Virus Filtration: Impact of Protein Concentration on LRV

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## Parvovirus LRV (Antibody Concentration)-20N

| IgG Conc. (mg/ml) | PPV LRV | Filtration Time (hr) | Filtration Volume (L/m²) | Throughput (kg/m²) |
|---|---|---|---|---|
| 1 | ≥ 5.67 | 6.4 | 350 | 0.35 |
| 5 | ≥ 5.37 | 6 | 280 | 1.40 |
| 10 | ≥ 6.00 | 6 | 240 | 2.40 |
| 20 | ≥ 5.50 | 6 | 148 | 2.96 |
| 30 | ≥ 5.58 | 6 | 98 | 2.94 |
| 50 | ≥ 5.67 | 5 | 24 | 1.20 |

100 mM NaCl, pH 4.5,
Pressure; 78.4 kPa,
Serum-free PPV 0.5 vol%

Connecting People, Science and Regulation

## Parvovirus LRV (Antibody Concentration)-BioEX

| IgG Conc. (mg/ml) | PPV LRV | Filtration Time (hr) | Filtration Volume (L/m²) | Throughput (kg/m²) |
|---|---|---|---|---|
| 1 | ≥ 5.42 | 4 | 490 | 0.49 |
| 5 | ≥ 5.78 | 4 | 470 | 2.35 |
| 10 | ≥ 5.35 | 4 | 425 | 4.25 |
| 30 | ≥ 5.28 | 6.4 | 300 | 9.00 |
| 50 | ≥ 5.10 | 9.4 | 180 | 9.00 |

100 mM NaCl, pH 4.5
Pressure; 294 kPa,
serum-free PPV 0.5 vol%

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*Tomoko Hongo-Hirasaki, 2011 PDA Virus & TSE Safety Conference* (adapted)

- ✓ No impact on PPV LRV (>4 log) over wide range of protein concentrations

# Robust Virus Filtration: Impact of pH on LRV

## Parvovirus LRV (pH) - 20N

| pH | PPV LRV | Filtration Time (hr) | Filtration Vol. (L/m$^2$) |
|---|---|---|---|
| 4.5 | ≥ 6.00 | 6.0 | 240 |
| 5 | ≥ 6.00 | 6.0 | 232 |
| 6 | ≥ 6.00 | 6.0 | 217 |
| 7 | ≥ 5.78 | 6.0 | 205 |
| 8 | ≥ 6.09 | 6.0 | 213 |

10 mg/ml IgG, 100 mM NaCl
Serum-free PPV 0.5 vol%,
Pressure; 78.4 kPa

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## Parvovirus LRV (pH) - BioEX

| pH | PPV LRV | Filtration Time (hr) | Filtration Vol. (L/m$^2$) |
|---|---|---|---|
| 4.5 | ≥ 5.28 | 6.4 | 300 |
| 5.6 | ≥ 5.56 | 9.3 | 215 |
| 6.7 | ≥ 5.14 | 9.0 | 150 |
| 7.6 | ≥ 5.40 | 9.6 | 155 |

30 mg/ml IgG, 100 mM NaCl
Serum-free PPV 0.5 vol%,
Pressure; 294 kPa

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*Tomoko Hongo-Hirasaki, 2011 PDA Virus & TSE Safety Conference* (adapted)

✓ No impact on PPV LRV (>4 log) over wide range of pH

# Robust Virus Filtration: Impact of Ionic Strength on LRV

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## Parvovirus LRV (Ionic Strength) - 20N

| NaCl Conc. (mM) | PPV LRV | Filtration Time (hr) | Filtration Volume (L/m$^2$) |
|---|---|---|---|
| 1 | ≥ 5.84 | 6.0 | 152 |
| 10 | ≥ 6.34 | 6.0 | 200 |
| 100 | ≥ 6.00 | 6.0 | 240 |
| 200 | ≥ 5.67 | 6.0 | 220 |
| 500 | ≥ 6.00 | 6.0 | 190 |

10 mg/ml IgG, pH 4.5
Serum-free PPV 0.5 vol%,
Pressure; 78.4 kPa

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## Parvovirus LRV (Ionic Strength) - BioEX

| NaCl Conc. (mM) | PPV LRV | Filtration Time (hr) | Filtration Vol. (L/m$^2$) |
|---|---|---|---|
| 50 | ≥ 5.48 | 6.0 | 300 |
| 100 | ≥ 5.28 | 6.0 | 300 |
| 250 | ≥ 5.28 | 9.5 | 280 |
| 500 | ≥ 5.92 | 9.5 | 160 |

30 mg/ml IgG, pH 4.5
Serum-free PPV 0.5 vol%,
Pressure; 294 kPa

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*Tomoko Hongo-Hirasaki, 2011 PDA Virus & TSE Safety Conference* (adapted)

✓ No impact on PPV LRV (>4 log) over wide range of ionic strength

# Robust Virus Filtration: Impact of Temperature on LRV

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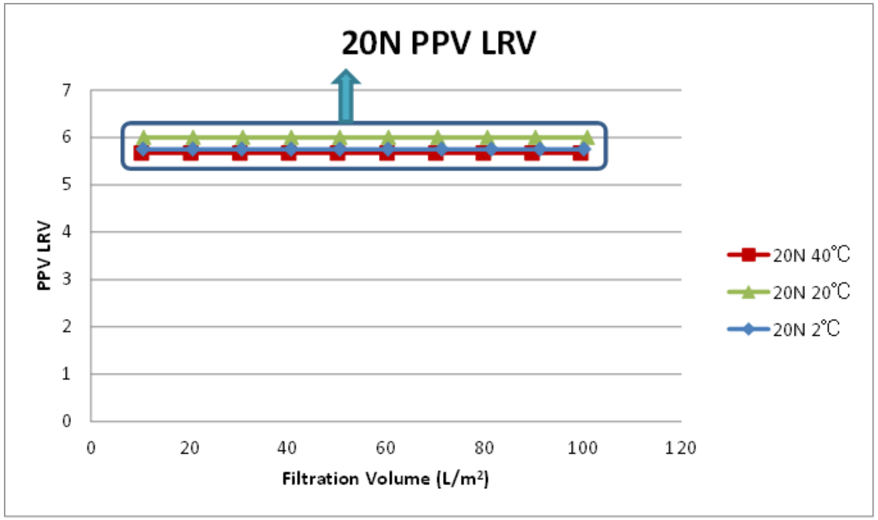
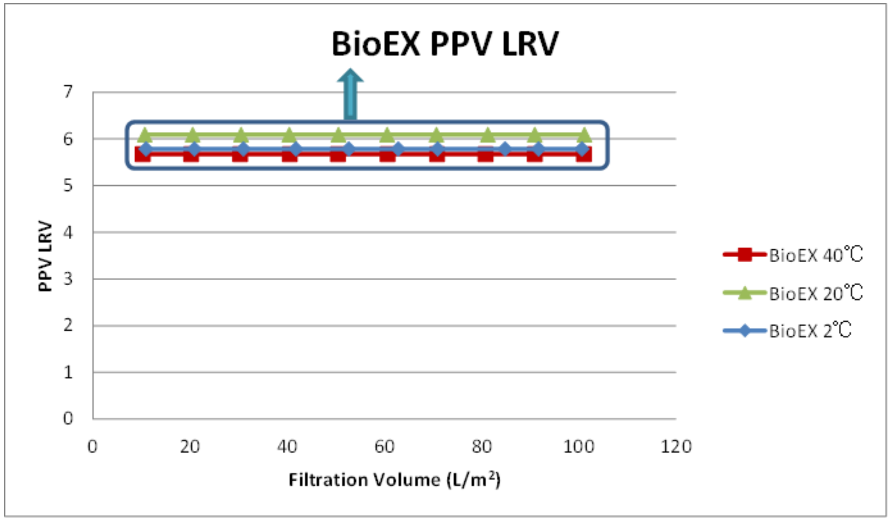



10 mg/ml IgG, 100 mM NaCl, pH 4.6
Serum-free PPV, 0.5 vol%
Pressure: 98 kPa

20 mg/ml IgG, 100 mM NaCl, pH 4.5
Serum-free PPV, 0.5 vol%
Pressure: 343 kPa

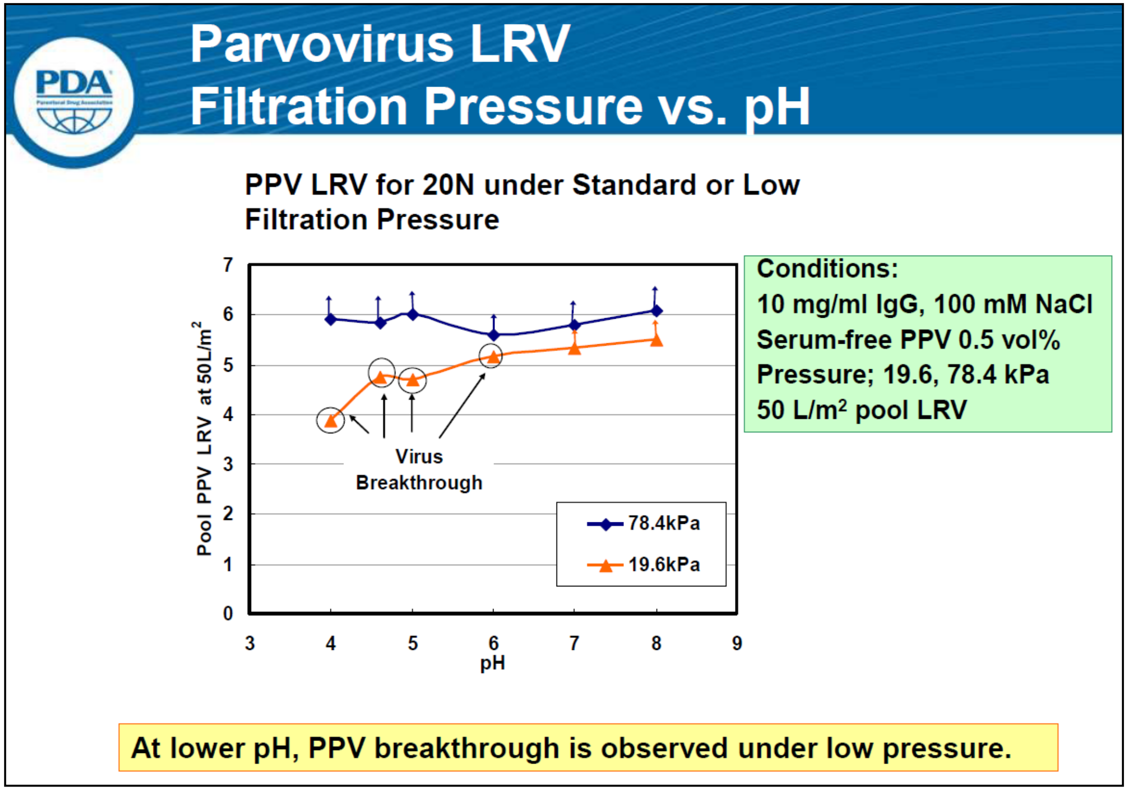
- ✓ No impact on PPV LRV (>4 log) over wide range of temperature



*Tomoko Hongo-Hirasaki, Tenth Plasma Product Biotechnology Meeting 2017*

# Impact of Filtration Pressure

- The effects of low pressure and depressurization are dependent on other factors




*2011 PDA Virus and TSE Safety, Tomoko Hongo-Hirasaki, Asahi Kasei Medical*

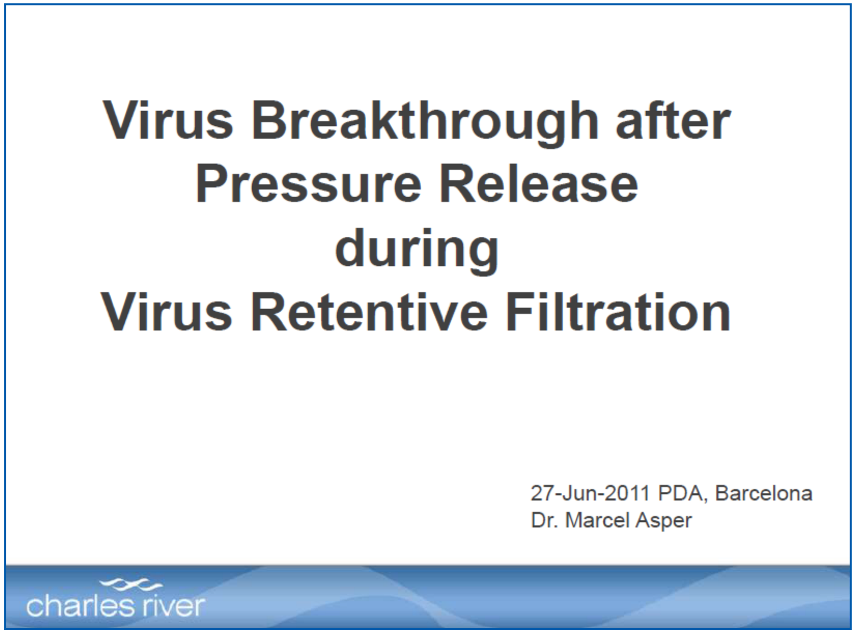
**Other factors:**

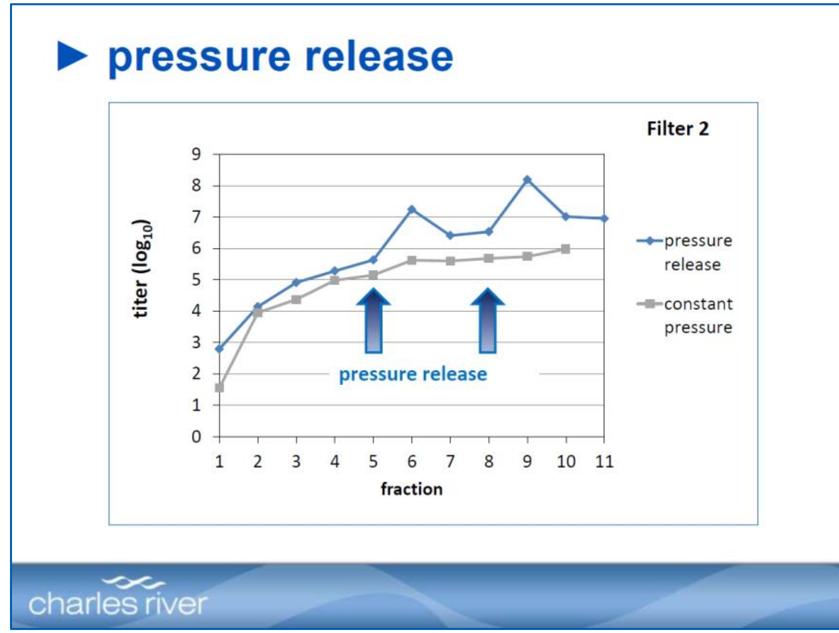
- pH
- NaCl concentration
- Product
- Virus challenge

# Process Pause Impact



- **In 2011, the impact of pressure release was presented at the PDA Virus and TSE Safety Meeting**




Most filter types display significant impact on <u>parvovirus</u> throughput due to the process pause

*2011 PDA Virus and TSE Safety, Marcel Asper, Charles River Laboratories* (adapted)

# Robust Virus Filtration - Summary

- Based on size exclusion mechanism, Planova filters demonstrated robust performance across broad factors;

  protein conc., ionic strength, pH, temperature

- Pressure control is very important for virus filtration.

  ↓

  We investigated:

  ✓ What is the additional mechanism to explain these phenomena?

  ✓ Design space study for robust and stable performance across broad ranges of parameters

# Overview

# Visualization of Virus Distribution Inside Membrane

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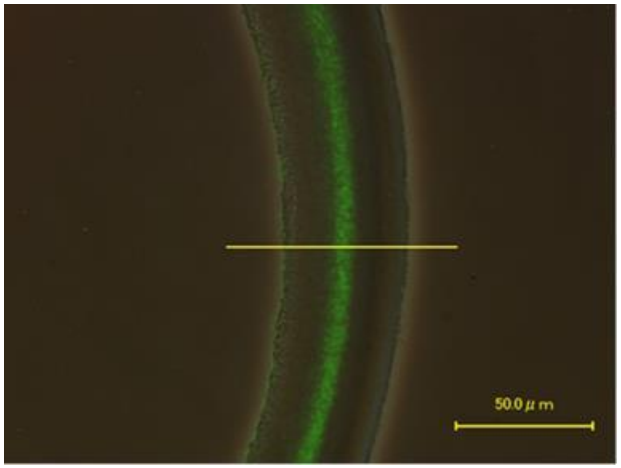
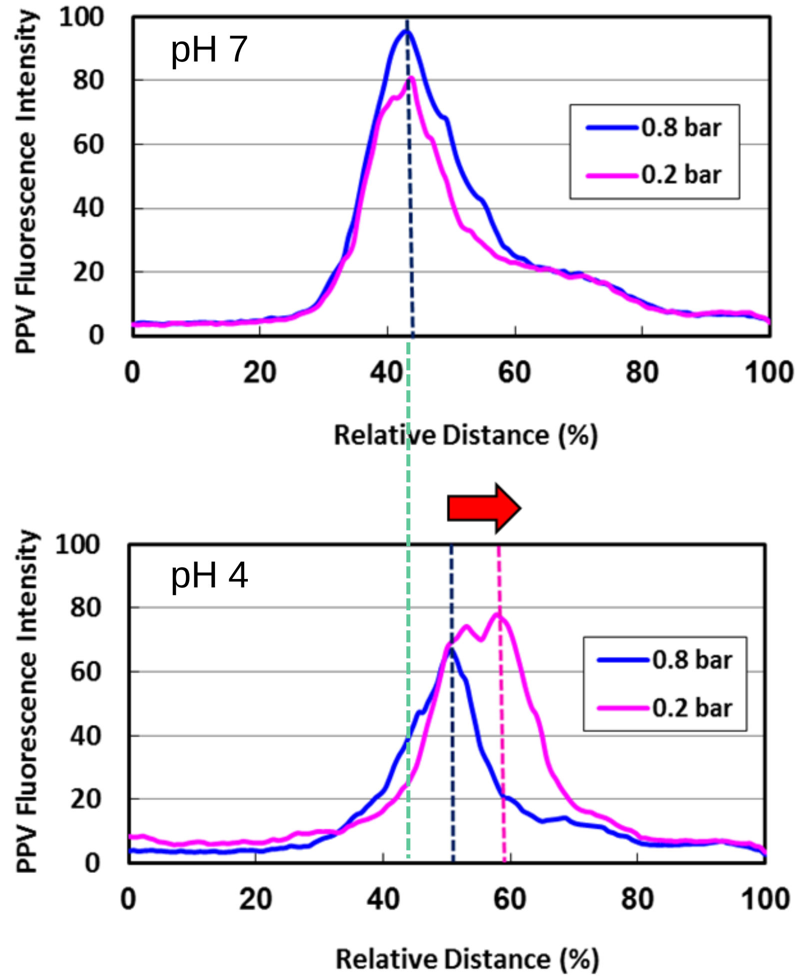
**Cross-sectional view of membrane using fluorescence microscopy**




**In** **Out**




✓ Small virus entrapment location may depend on factors such as solution conditions and filtration pressure.



*Tomoko Hongo-Hirasaki, 2014 Planova Workshop*

# Additional Mechanism: Interactions

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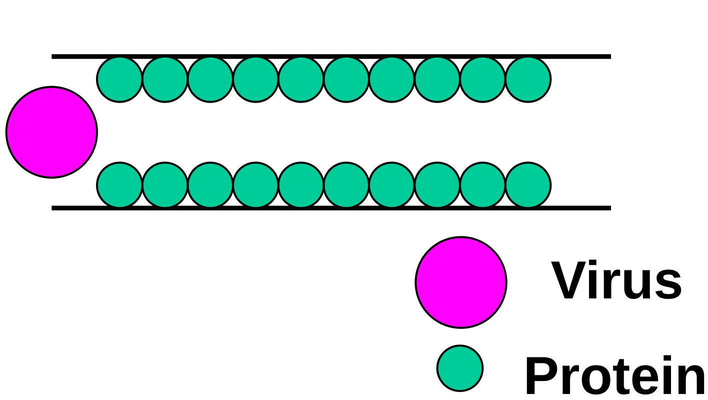
- ❑ **Physicochemical properties (solution condition and/or molecule type) may impact the size-exclusion mechanism**

**Two hypothetical cases**

**Pore size reduction**

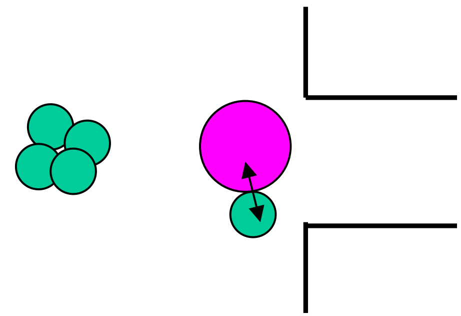



Pore size reduction by adsorption of protein on membrane

Increase in virus removal capability

Decrease in flux

**Aggregation**



Increase in interactions, complexes act as large particles

Increase in virus removal capability

Decrease in flux

✓ Possible interactions: surface charge, or hydrophobicity of virus, protein and membrane

*Adapted from Tomoko Hongo-Hirasaki, 2013 PDA Virus & TSE safety*

# Additional Mechanism: Mechanistic Effect

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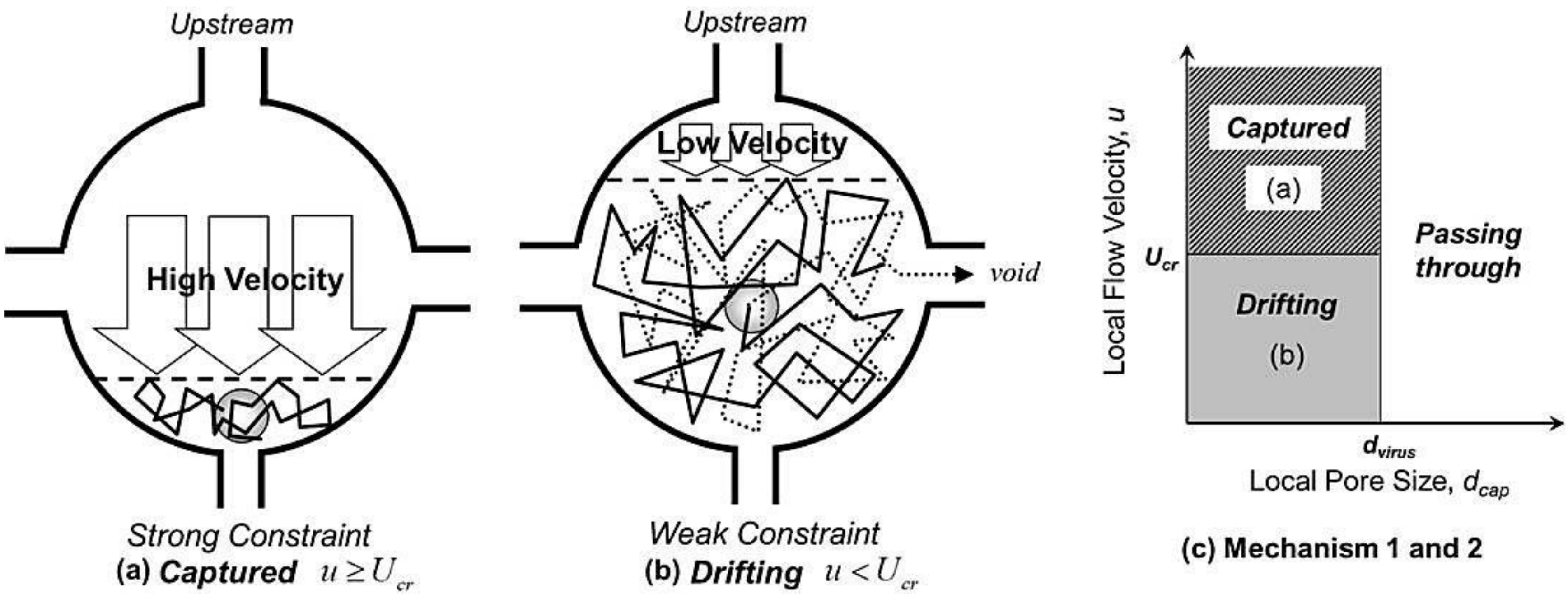
- **Constraint by hydrodynamic force**

  Hydrodynamic force controls Brownian motion of viruses




***u***, **Flow velocity is controlled by filtration pressure**
***$U_{cr}$***, **Critical velocity to overcome Brownian motion**

**Mechanism 1: Size exclusion $d_{virus} > d_{cap}$**
**Mechanism 2: Hydrodynamic force**



*A Yamamoto, T Hongo-Hirasaki, Y Uchi, H Hayashida and F Nagoya. AIChE Journal, 2014, 60(6): 2286–2297*

# Design Space Study for Parvovirus LRV

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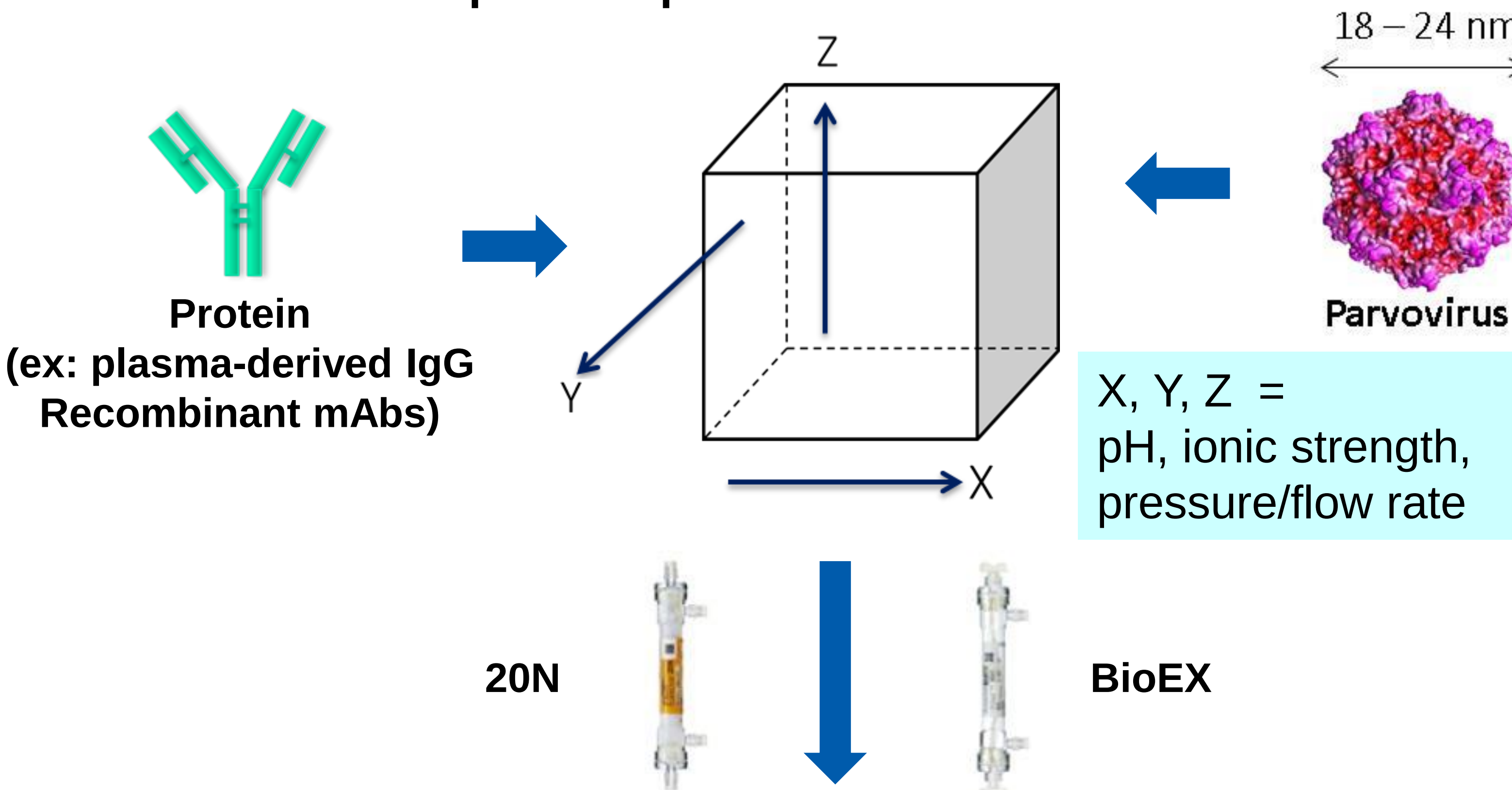
- **Performance and process parameters**




- Study goals: Define the robust operating spaces for Planova filters
  - ✓ Design spaces for Planova 20N and BioEX filters which provide highly effective parvovirus removal (LRV ≥ 4)

# Design Space Study:
# Design Space of PPV LRV

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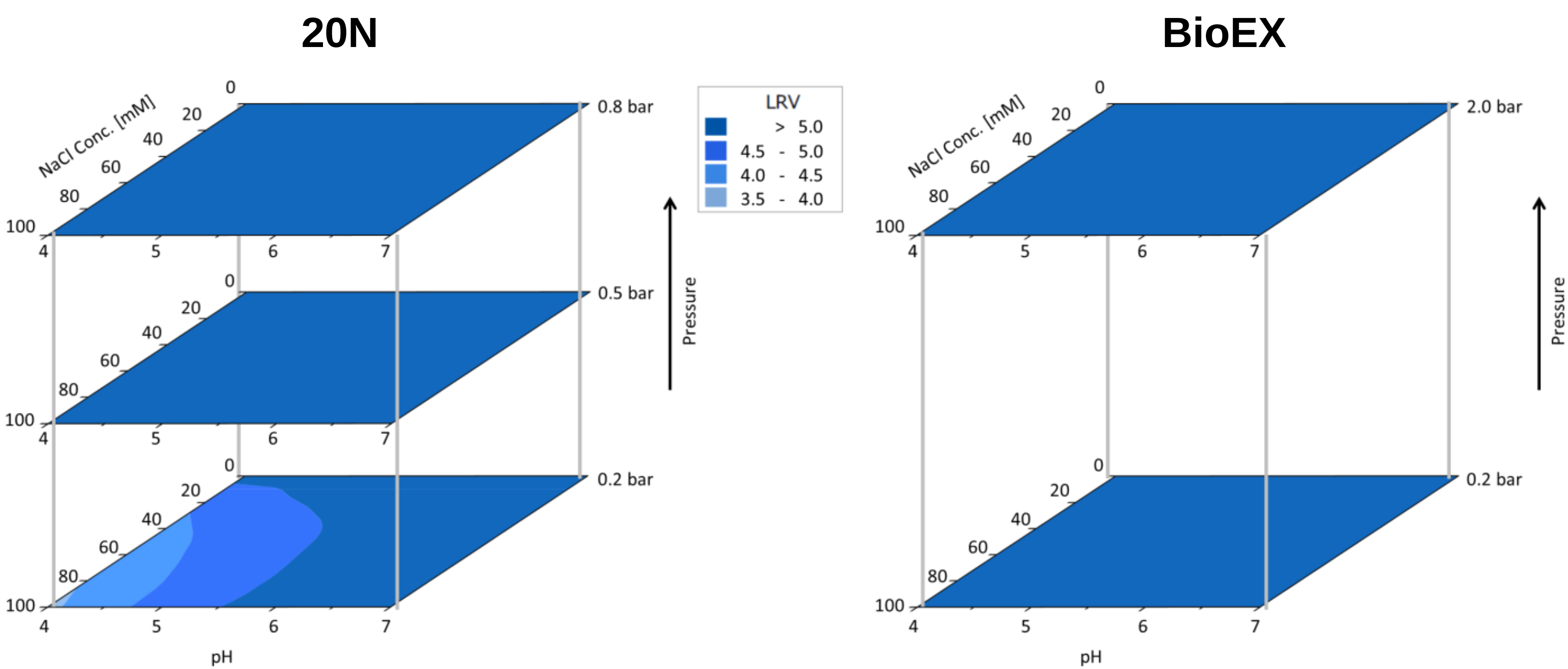
## Human IgG




✓ Planova 20N and BioEX showed robust performance over wide range of conditions in design space studies.



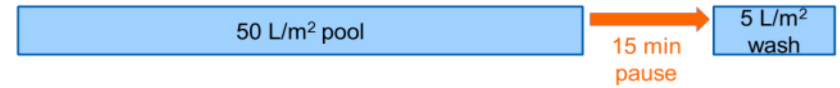
*Tomoko Hongo-Hirasaki, Tenth Plasma Product Biotechnology Meeting 2017*

# Planova Design Space Collaboration - Janssen

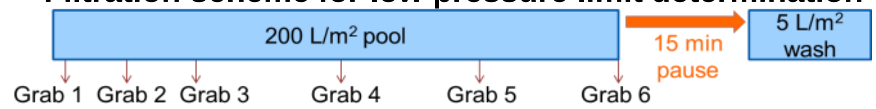
## Collaboration with Janssen

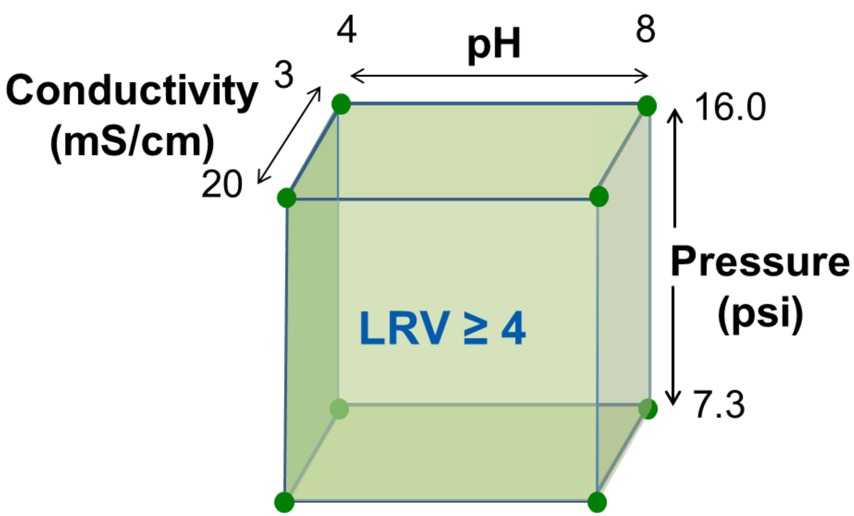
- Three mAb products (IgG1)
- Spiked with MVM (~6 log $TCID_{50}$/mL)

**Filtration scheme for initial scouting and design space**

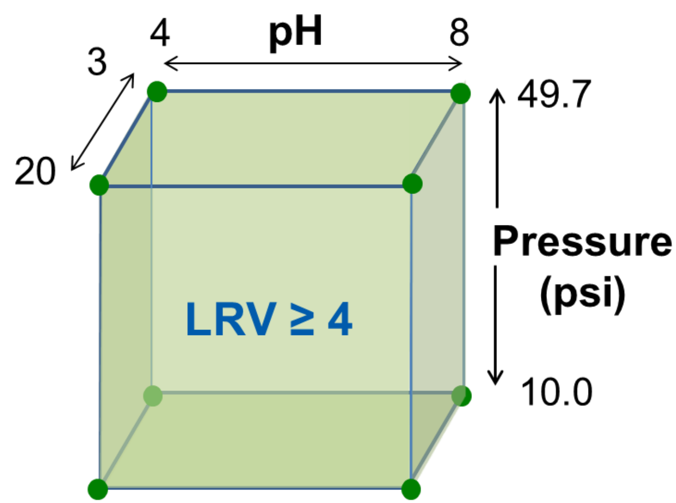



**Filtration scheme for low pressure limit determination**




**Planova 20N**




**Planova BioEX**

For Planova 20N and BioEX, operating pressures greater than 7.3 and 10.0 psi (0.50 and 0.69 bar), respectively, are expected to provide LRV greater than 4 for most mAb processes

# Implementation in DSP

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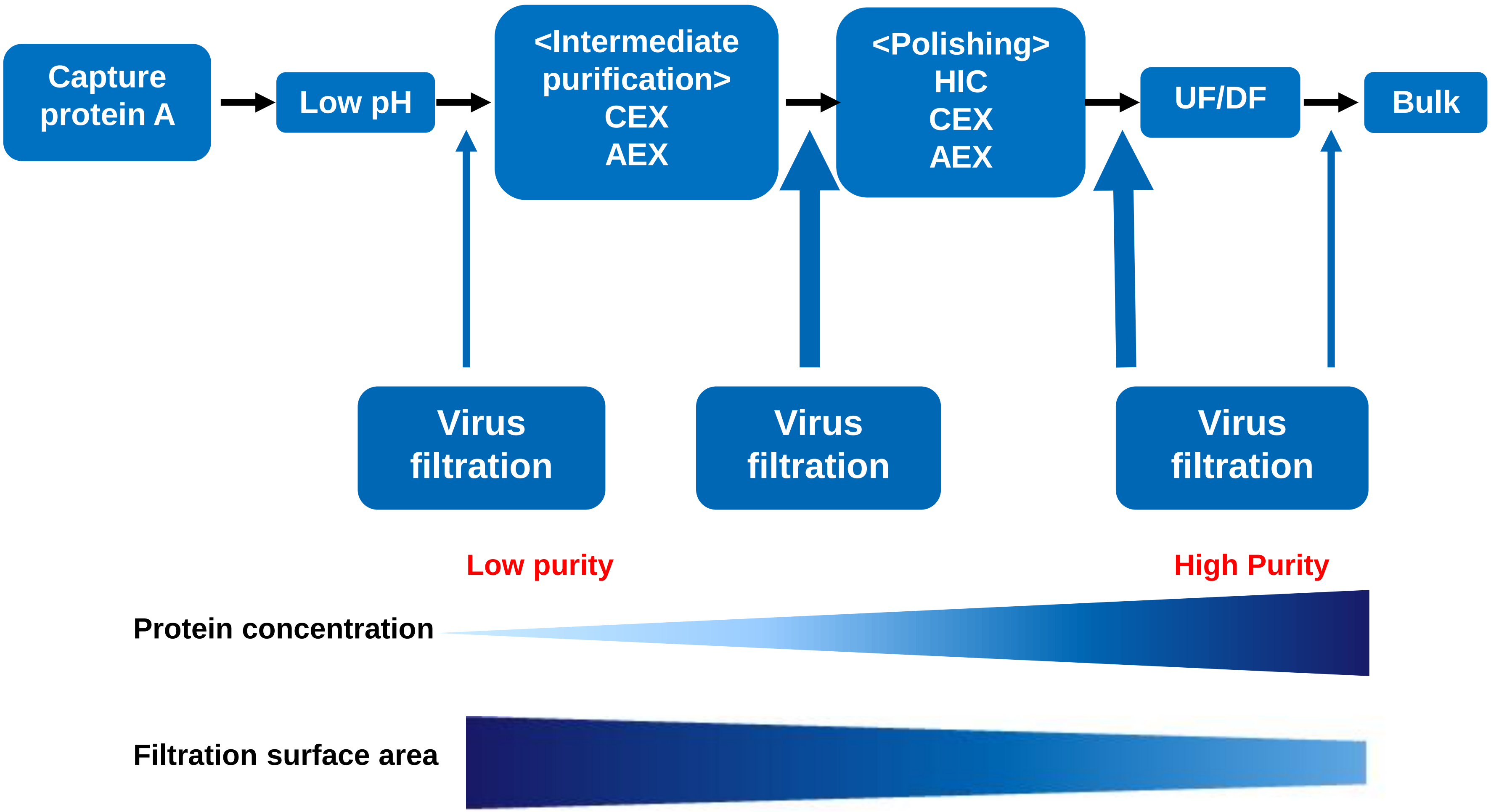


- ✓ Flexible process optimization possible with placement of virus filtration step
- ✓ Platform possible with applicability to wide range of conditions

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# Considerations for Viral Clearance Studies

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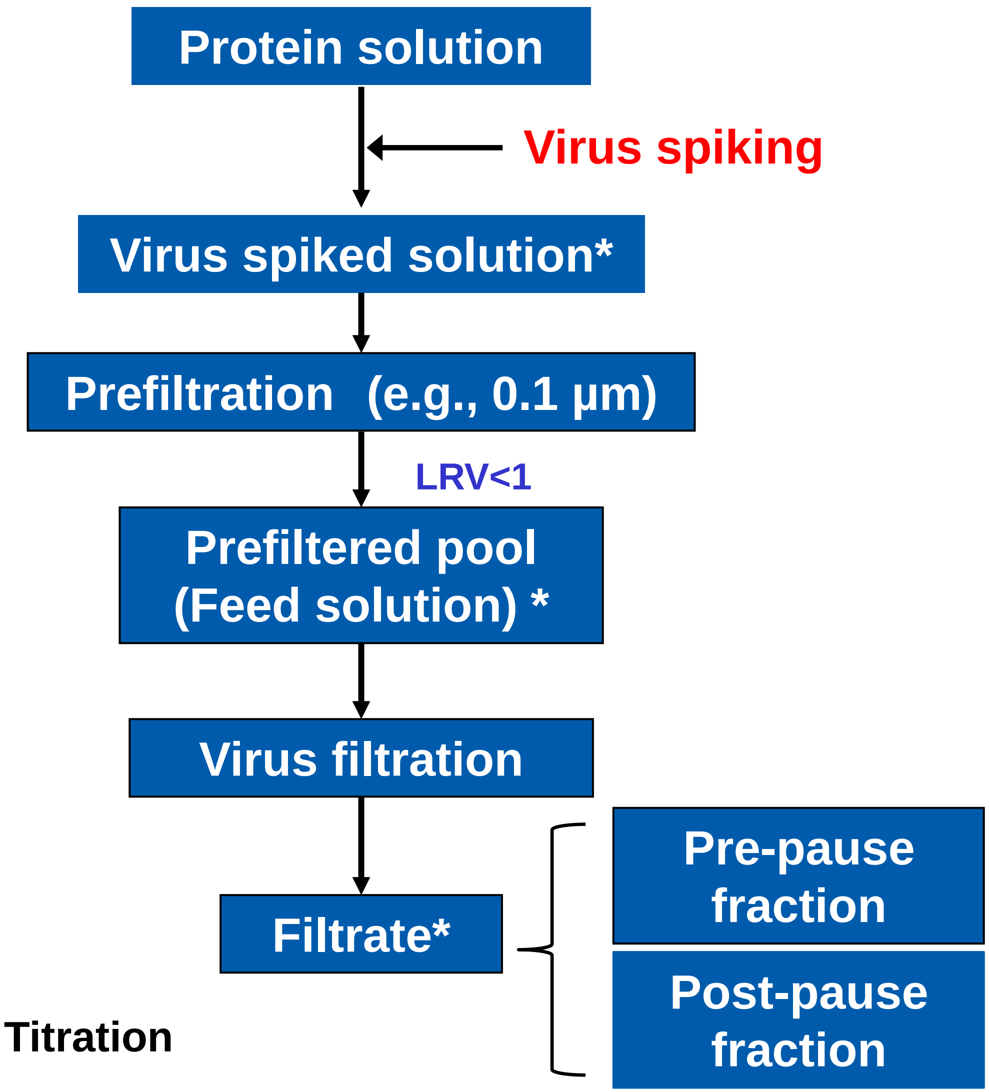
- ❑ **Procedure for viral clearance study**




* Titration

# Virus Trapping Capacity

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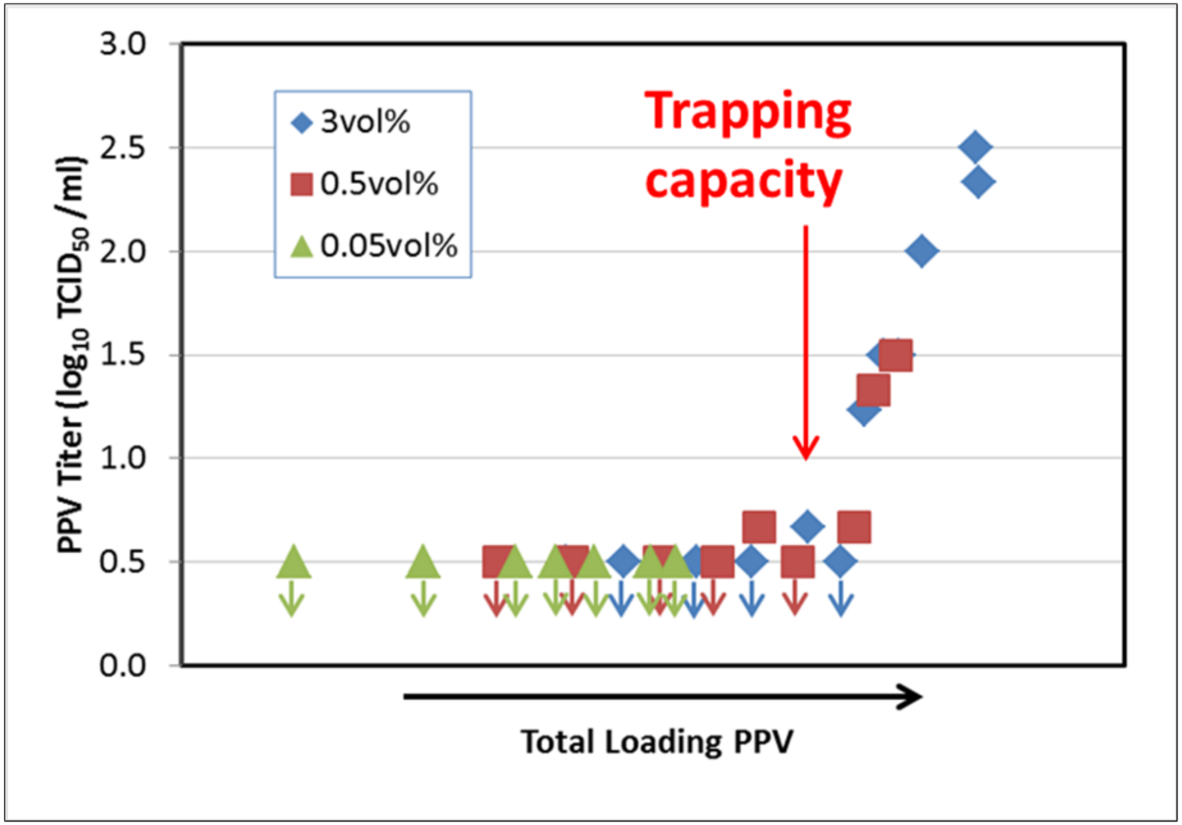
- ❑ **Virus filters have a finite virus capture capacity**




10 mg/mL human IgG

Serum-free PPV spiking:
0.05, 0.5, 3.0 vol%

*Tomoko Hongo-Hirasaki, 2014 Planova Workshop* (adapted)

# Impacts of Overspiking Virus

- CRO experience:

| Protein /run | Logs virus loaded (PFU) | LRV filtrate pool | Comments |
|---|---|---|---|
| A1 | 9.2 | ≥7.6 | Inconsistent clearance between replicates |
| A2 | 9.3 | 6.1 | |
| B1 | 9.2 | ≥7.4 | |
| B2 | 9.1 | 5.8 | |
| C1 | 9.4 | 3.2 | Low overall reduction |
| C2 | 9.5 | 3.3 | |
| D1 | 9.0 | 3.1 | |
| D2 | 8.9 | 3.4 | |

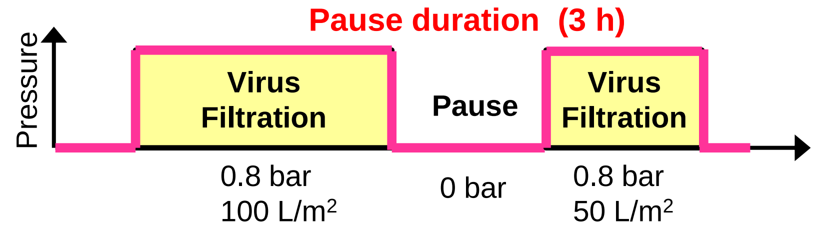
**Observation**

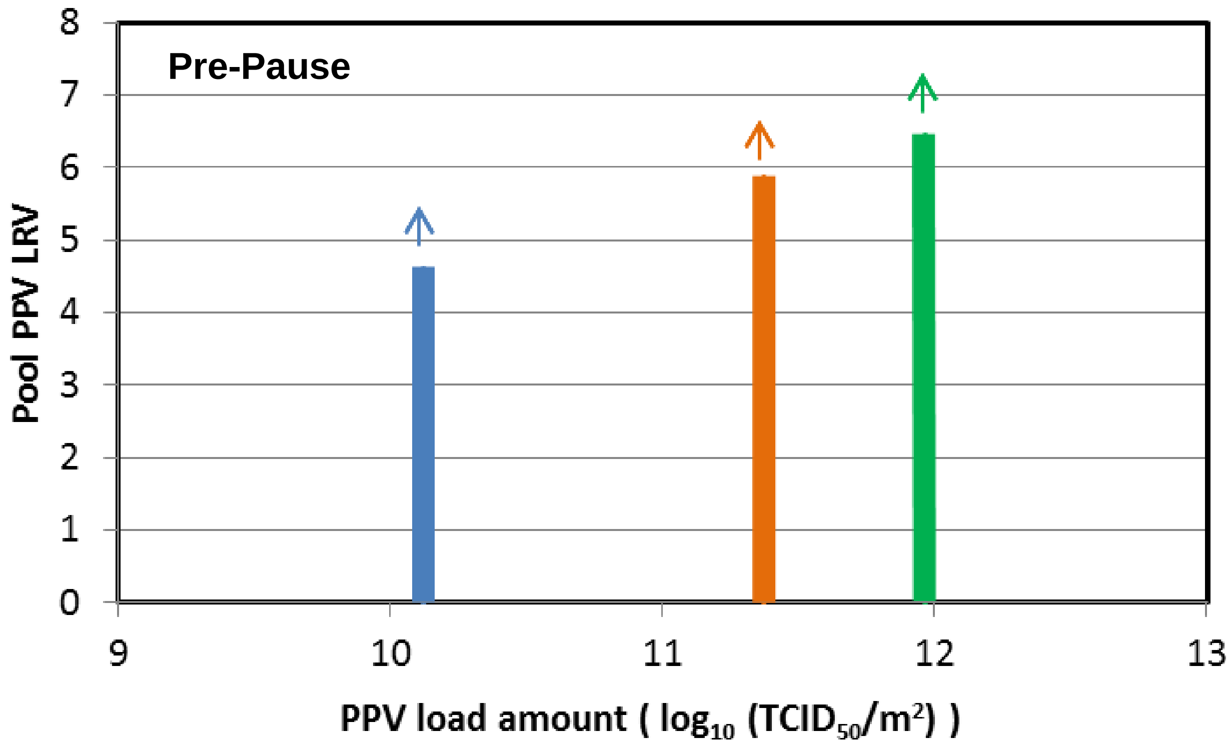
- On filters from multiple suppliers, overspiking can lead to inconsistent results between duplicate runs or low parvovirus clearance

*Mike Burnham, WuXi AppTec, 2016 Planova Workshop* (adapted)

# Viral Clearance Studies: Impact of Virus Load on LRV

Planova 20N
10 mg/mL h-IgG; 0.8 bar; pH 7, 100 mM NaCl
Virus spike: 0.03, 0.3, 3.0 vol%




✓ Pre-pause PPV LRV increases with virus load.

# Viral Clearance Studies: Impact of Virus Load on LRV

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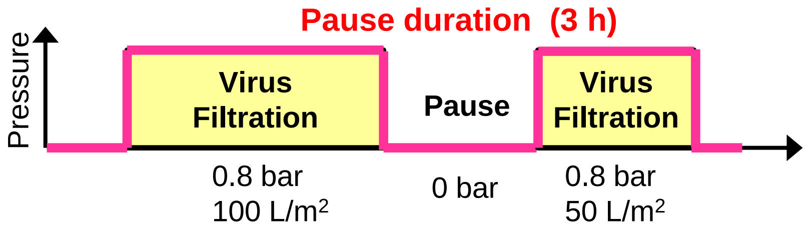
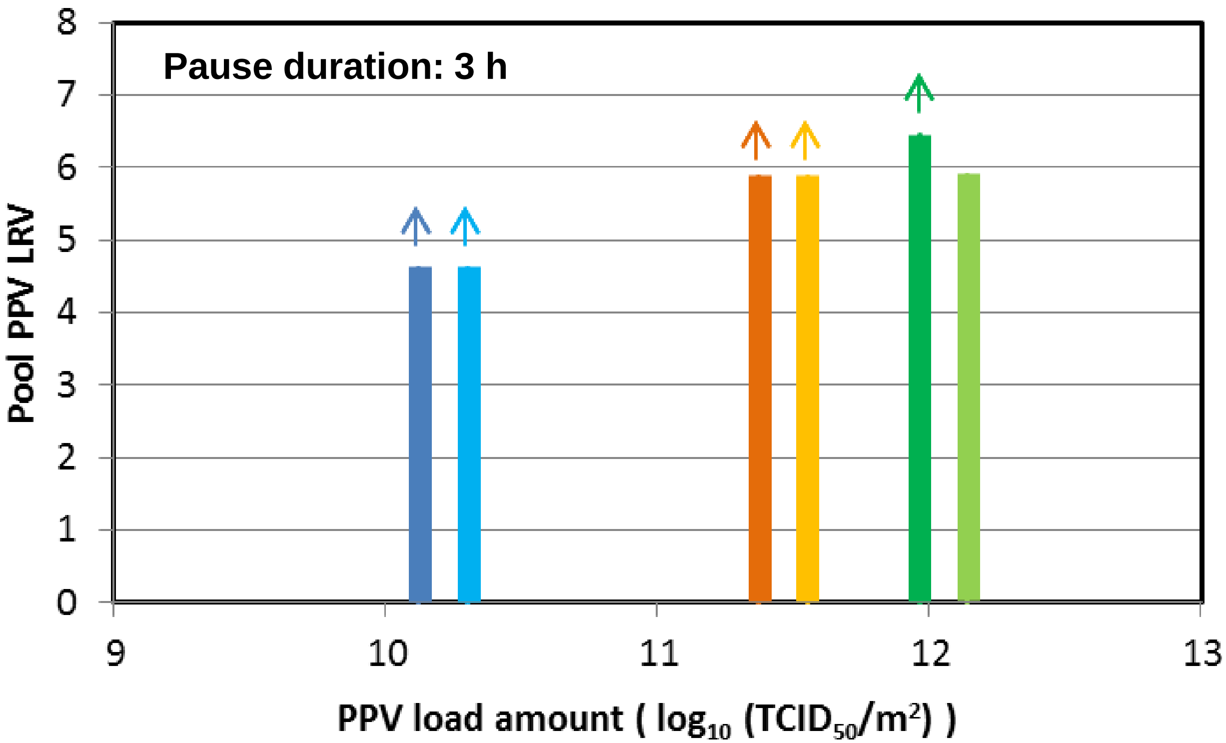
Planova 20N
10 mg/mL h-IgG; 0.8 bar; pH 7, 100 mM NaCl
Virus spike: 0.03, 0.3, 3.0 vol%




- ✓ Pre-pause PPV LRV increases with virus load.
- ✓ It is important to consider the optimal balance of target LRV and virus load.
- ✓ Select appropriate virus load to obtain the target LRV.

# Consideration for Virus Load

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- The virus load solution contains a high number of non-infectious virus particle content in addition to infectious particles.

- The total virus particle number could be more than expected and it would sometimes cause unreliable LRV.

- Ratio of non-infectious to infectious particles is variable for parvovirus (e.g., MVM): c.a., 250:1 – 2,500:1

  Sampling of references:
  - ✓ *Virology*, 2015, 481: 63-72
  - ✓ *J. Virol.,* 2005, 79: 289-98
  - ✓ *J. Virol.*, 2010, 84(15):7782-92

- In consideration of the limited virus capture capacity of virus filters, selection of virus load sufficient to achieve >4 log reduction is important.

# Optimizing Viral Clearance Studies

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- **Select appropriate virus load in virus filtration studies**
  - Spiking appropriately can minimize assay artifacts
  - We recommend not to use spiking based on percentage ratio
  - Spike to a targeted total viral challenge
  - Typically, spiking to a total virus challenge of $\boxed{10.5\text{-}11.0 \text{ log IU/m}^2}$ is sufficient to obtain a virus LRV of 5-6

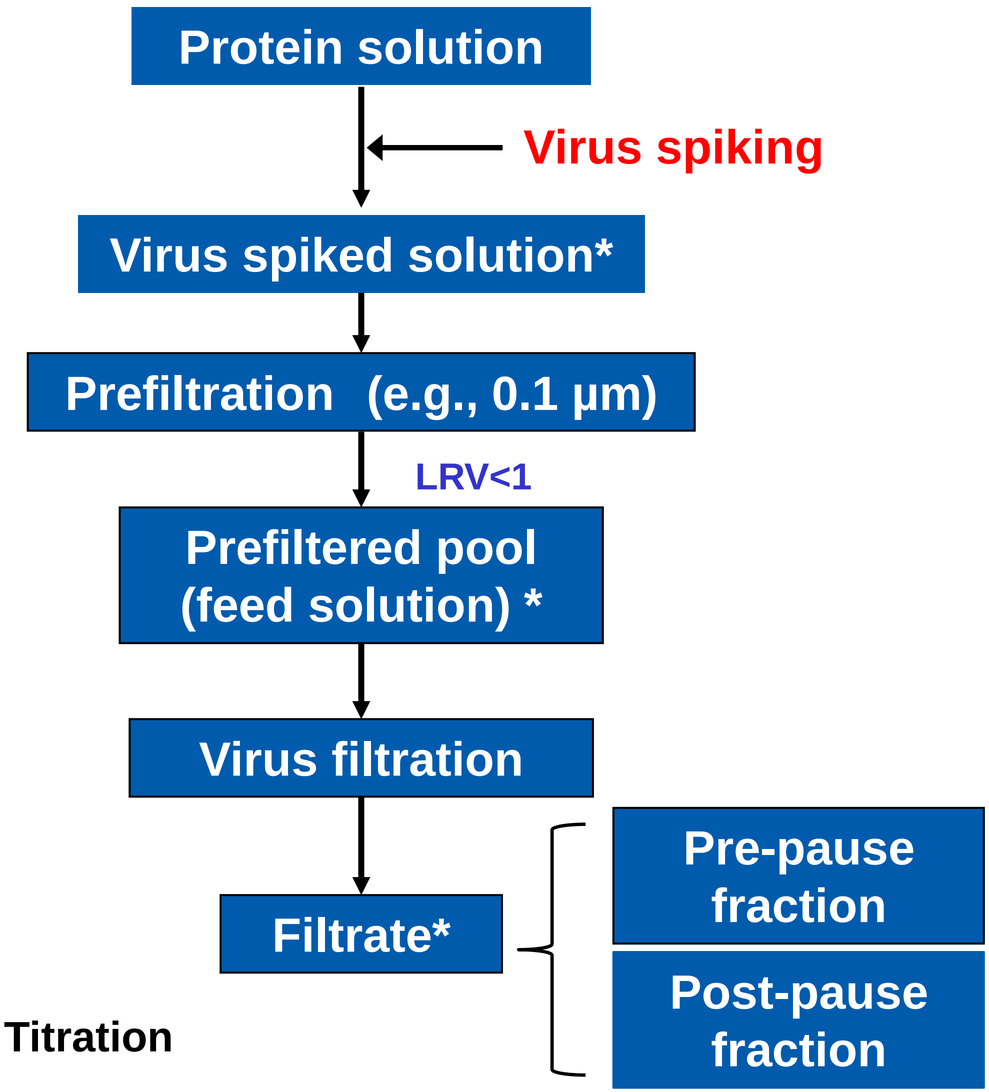
7.5–8.0 log IU on a 10 $cm^2$ filter

**Example:**

| Method | 1% spike ratio | Target challenge 8.0 log IU |
|---|---|---|
| **Load volume** | 300 mL | 300 mL |
| **Virus stock titer** | 9 log IU/mL | 9 log IU/mL |
| **Spike volume** | 3 mL | 0.1 mL |
| **Viral load** | 7.0 log IU /mL | 5.5 log IU/mL |
| **Total viral load / 10cm²** | 9.5 log IU | 8.0 log IU |

# Considerations for Viral Clearance Studies

❑ **Procedure for viral clearance study**




*** Titration**

Points

✓ Select appropriate virus load to obtain the target LRV

✓ Plan at least the same process pause duration as in manufacturing process (e.g., for post-wash)

# Overview

1. Introduction
2. Findings for Planova Virus Filtration
   2.1. Key Factors of Robust Virus Filtration Processes
   2.2. Design Space for Robust and Stable Performance
3. Findings for Appropriate Viral Clearance Studies
4. **Conclusions**

# Conclusion

- **To develop robust virus filtration processes**

### 1. Planova virus filtration

- ✓ Based on size exclusion mechanism, Planova filters demonstrate robust small virus removal and filterability across broad conditions.
- ✓ Other mechanisms (factors) must be considered on a case-by-case basis (each product and process) in order to achieve higher small virus removal capability.
- ✓ Planova filters shows robust performance over wide range of conditions in design space studies.

### 2. Evaluation by appropriately designed viral clearance study

Selection of appropriate virus load and experimental design can improve outcomes and avoid study artifacts in viral clearance studies.

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